IOT Project Proposal

Team Members

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Track Choice

Track B — Computer Vision IoT System

Project Title

Smart Security Camera for Verified User Access

Problem Statement

Home security systems often rely on motion sensors or alarms that cannot distinguish between trusted residents and strangers. This can lead to false alarms, unnecessary alerts, and reduced usability. Our system addresses the need for a smart, automated method to verify authorized individuals and detect intruders in real time.

By combining IoT sensors and an AI-powered camera, the system can register trusted users and verify their identity dynamically, improving security while minimizing false alarms. This is an IoT problem because it involves connected devices, sensor-triggered communication, and remote processing to make real-time decisions.

Short System Idea:

Our system consists of:

Device Layer:
AI-enabled camera
Motion/proximity sensor
The camera captures faces and can also be triggered by the REST API during user registration

Gateway Layer:
IoT gateway (e.g., Raspberry Pi or microcontroller)
Collects device events and coordinates communication between devices, MQTT, and the REST API
Forwards real-time alert and event data to MQTT
Interacts with the REST API for private operations such as registration, verification, and detection control

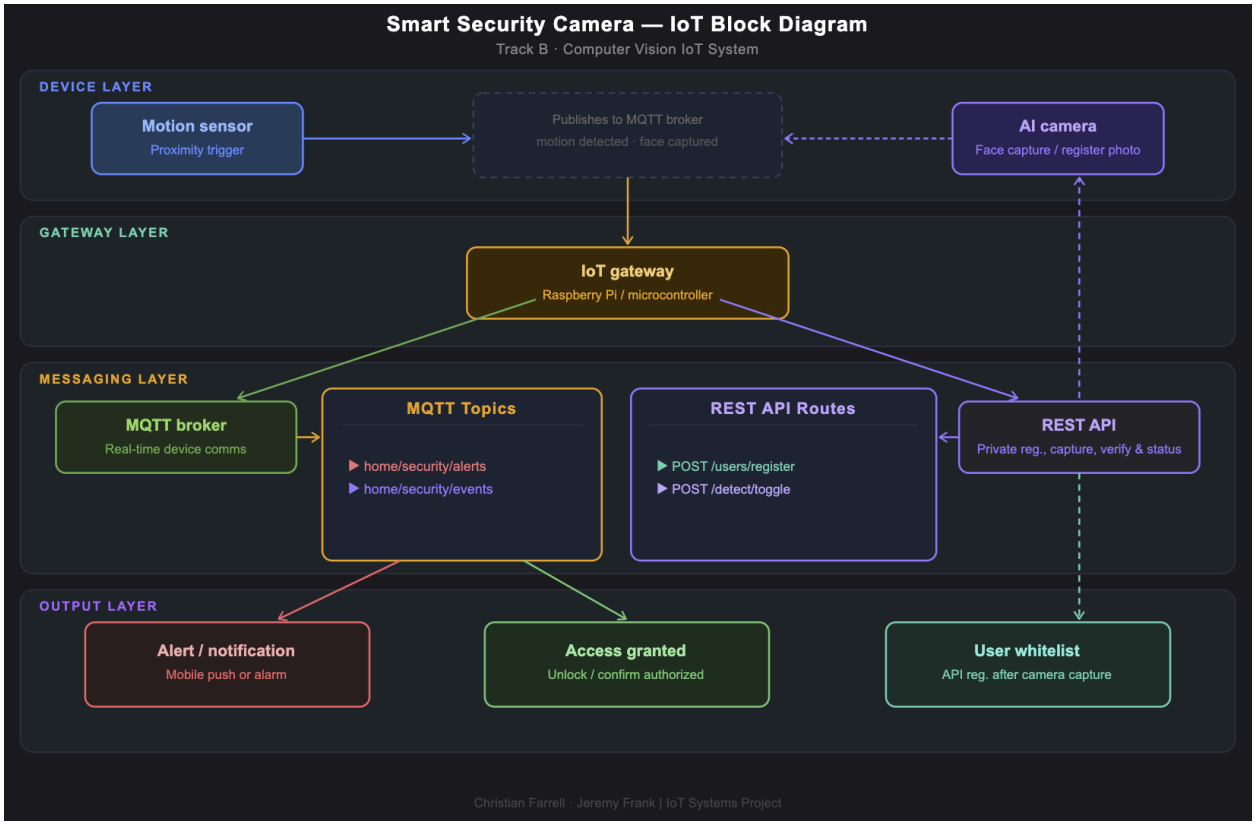
Messaging Layer:
MQTT broker for real-time event messaging only

home/security/alerts → alert published when an unknown user is detected
home/security/events → general motion/detection event stream

REST API for private management and control:
POST /users/register → requests camera capture for a new user, then stores that user in the whitelist
POST /detect/toggle → enables or disables detection behavior
Private API operations are used for registration, verification-related control, and system status rather than public MQTT topics

Output Layer:
Whitelist of registered users stored securely on the server
Access granted action for authorized users
Alerts or actions triggered for unverified individuals (e.g., mobile notification, alarm)

Block Diagram:



Scope Statement:

The project is realistic for the semester as it involves:

Using off-the-shelf sensors and AI camera modules

Implementing a simple REST API for register and verify commands
Using MQTT for relaying sensor data to topics
Storing and verifying a small whitelist of users locally or on a lightweight server
Building a basic alert or notification system

All components can be prototyped with Python, OpenCV, and a small IoT gateway, keeping the system manageable and achievable within the semester timeline.